

Tae-Geun Kim

Postdoctoral Researcher

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Appointments

Fudan–RIKEN Joint Postdoctoral Fellow

Nov. 2025 – Present

Key Laboratory of Nuclear Physics and Ion-beam Application (MOE), Fudan University, Shanghai, China;
RIKEN Center for Interdisciplinary Theoretical and Mathematical Sciences (iTHEMS), Wako, Japan

Education

Ph.D. in Theoretical Particle Physics, Yonsei University, Seoul, Republic of Korea

Mar. 2017 – Aug. 2025

Thesis: *Intercommunication between Artificial Intelligence and Physics* (Supervisor: Prof. Seong Chan Park)

B.Sc. in Astronomy, Yonsei University, Seoul, Republic of Korea

Mar. 2012 – Feb. 2017

Military Service

Technical Research Personnel, Republic of Korea Army (ROKA)

Sep. 2022 - Aug. 2025

Research Areas & Expertise

• Dark matter physics

- ▶ Phenomenology of axion-like particles & primordial black holes
- ▶ Detectability studies of dark matter candidates using various astrophysical and cosmological probes
- ▶ Theoretical modeling and simulation of dark matter interactions

• Machine Learning for Physics

- ▶ Development of deep learning models for missing information search and anomaly detection in high-energy physics data
- ▶ Neural network approaches to learn and emulate complex physical systems and dynamics
- ▶ Functional and operator learning for solving differential equations and modeling physical phenomena

• Scientific & High Performance Computing

- ▶ Design and implementation of efficient numerical algorithms for physics simulations
- ▶ Optimization of computational methods for large-scale data analysis in astrophysics and particle physics
- ▶ Development of high-performance software tools for scientific computing, with a focus on Rust-based solutions
- ▶ Application of parallel computing techniques to accelerate physics computations

Programming Skills & Tools

• Primary Languages: Rust, Python, C++, Julia

• Frameworks & Libraries

- ▶ **Numerical Computing:** `peroxide`, `numpy`, `scipy`, `pandas/polars`, `BLAS/LAPACK`, `eigen`, `matlab`, `mathematica`
- ▶ **Machine Learning:** `pytorch`, `jax/equinox/optax`, `wandb`, `optuna`, `candle`, `tensorflow`, `scikit-learn`
- ▶ **Visualization:** `matplotlib`, `vegas`, `ggplot2`, `plotly`
- ▶ **High Energy Physics:** `BlackHawk`, `galprop`, `madgraph`, `root`
- ▶ **Quantum Computing:** `pennylane`, `qiskit`, `cirq`, `rustqip`
- ▶ **Web:** `django`, `vue`, `firebase`, `hugo`, `zola`, `elm`

Honors & Fellowships

- APS DCOMP Travel/Registration Award, APS Division of Computational Physics, CCP2026 (2026)
- Shanghai Superpostdoc Fellowship, Shanghai Municipal Government (2025-2027)
- Fudan Superpostdoc Fellowship, Fudan University (2025-2027)
- Academy Research Fellowship, Yonsei University (2022-2023)
- Best Oral Presentation Award, KPS 70th Anniversary and 2022 Fall Meeting (2022)

Publications

- 2026 *“Peroxide: A Batteries-Included Numerical Computing Library for Rust”*, **T.-G. Kim**, G. Comitini, J. Grage, B. Joens, M. Schreiber, et al., *J. Open Source Software* **11**, 10366
- 2026 *“Primordial Black Holes as a Factory of Axions: Extragalactic Photons from Axions”*, Y. Jho*, **T.-G. Kim***, J.-C. Park*, S. C. Park*, Y. Park*, *Prog. Theor. Exp. Phys.* **2026**, 023B07, [2212.11977]
- 2025 *“Learning Unknown Gravitational Fields for Spacecraft Trajectory Prediction and Estimation”*, T. Kim*, **T.-G. Kim***, A. Girard, I. Kolmanovsky, Under Review, [2501.18808]
- 2024 *“Neural Hamilton: Can A.I. Understand Hamiltonian Mechanics?”*, **T.-G. Kim**, S. C. Park, [2410.20951]
- 2024 *“HyperbolicLR: Epoch insensitive learning rate scheduler”*, **T.-G. Kim**, [2407.15200]
- 2024 *“Unsupervised sequence-to-sequence learning for automatic signal quality assessment in multi-channel electrical impedance-based hemodynamic monitoring”*, C. M. Hyun*, **T.-G. Kim***, K. Lee, *Comput. Meth. Prog. Bio.* **245**, 108079, [2305.09368]
- 2023 *“Missing information search with Deep Learning for Mass estimation (DeeLeMa)”*, K. Ban*, D. W. Kang*, **T.-G. Kim***, S. C. Park*, Y. Park*, *Phys. Rev. Research* **5**, 043186, [2212.12836]

* Co-first authors.

Invited Talks

- 2026 *“Can you trust your neural sampler?”* at CCP2026 @ Sejong University, Seoul, Korea
- 2026 *“Can one see the mass of a PBH?”* at KIAS Seminar, Seoul, Korea
- 2026 *“Neural Inverse Problems and Degeneracy: SMEFT and PBHs”* at PARTIQAL @ IBS, Daejeon, Korea
- 2026 *“Inverse Problems in High Energy Physics”* at DEEP-IN Starter Meeting @ RIKEN, Wako, Japan
- 2025 *“From Primordial Black Holes to Nuclei: Solving Inverse Problems in Physics with Operator Learning”* at Nuclear Lunch Seminar @ Fudan University, Shanghai, China
- 2025 *“PBH Phenomenology with Operator Learning”* at LSSU Seminar @ JBNU, Jeonju, Korea
- 2025 *“Can A.I. Understand Hamiltonian Mechanics?”* at DEEP-IN Seminar @ RIKEN, Online
- 2024 *“Can A.I. Understand Hamiltonian Mechanics?”* at KIAS, Seoul, Korea

Contributed Talks

- 2026 *“Can you trust your neural samplers?”* at *AI Agents for Nuclear Physics @ Fudan U.*, Shanghai, China
- 2026 *“Accelerating PBH Phenomenology via Neural Operators”* at *The 2nd AI+HEP in East Asia @ KEK*, Tsukuba, Japan
- 2025 *“Can AI Understand Hamiltonian Mechanics?”* at *29th International Summer Institute on Phenomenology of Elementary Particle Physics and Cosmology*, Yeosu, Korea
- 2025 *“A Neural Operator for Primordial Black Hole Physics”* at *The 4th workshop on Symmetry and Structure of the Universe*, Jeonju, Korea
- 2025 *“AI with Hamiltonian Mechanics: From Predictions to Understanding”* at *AI+HEP in East Asia 2025*, Daejeon, Korea
- 2025 *“Can A.I. Understand Hamiltonian Mechanics?”* at *Dark Matter as a portal to New Physics 2025*, Pohang, Korea
- 2025 *“Can A.I. Understand Hamiltonian Mechanics?”* at *Focused workshop on AI in High Energy Physics 2025*, Seoul, Korea
- 2024 *“Can A.I. Understand Hamiltonian Mechanics?”* at *2024 Korea-France STAR Workshop*, Seoul, Korea
- 2024 *“Can A.I. Understand Hamiltonian Mechanics?”* at *21st Saga-Yonsei Joint Workshop*, Seoul, Korea
- 2024 *“Primordial Black Hole dominant Axion background”* at *2024 KPS Fall Meeting*, Yeosu, Korea
- 2023 *“Exploration of PBHs and ALPs through a novel decay model on cosmological scale”* at *27th International Summer Institute on Phenomenology of Elementary Particle Physics and Cosmology*, Nantou, Taiwan

Teaching Experience

2025	Spring	Lecturer	Quantum Universe - Special Research , PHY 9901, Yonsei University, Seoul, Republic of Korea (Lecture about Deep Learning for Physics)
2021	Spring	TA	Introduction to General Relativity , PHY4208, Yonsei University, Seoul, Republic of Korea
2020	Fall	TA	Quantum Mechanics (2) , PHY3102, Yonsei University, Seoul, Republic of Korea
	Spring	TA	Quantum Mechanics (1) , PHY3101, Yonsei University, Seoul, Republic of Korea
2019	Fall	TA	Quantum Mechanics (2) , PHY3102, Yonsei University, Seoul, Republic of Korea
	Spring	TA	Elementary Particle Physics I , PHY8050, Yonsei University, Seoul, Republic of Korea
2017	Fall	TA	Gravity I: General Relativity , PHY8030, Yonsei University, Seoul, Republic of Korea
	Spring	TA	Mathematical Physics (1) , PHY4205, Yonsei University, Seoul, Republic of Korea

Selected Open Source Projects

Peroxide	Sep. 2018 – Present
• Comprehensive Rust numeric library for linear algebra, numerical analysis, and statistics	☆ : 723
• Customizable features for pure Rust, BLAS/LAPACK integration, and plotting capabilities	↓ : 1,250,135
• Includes automatic differentiation, special functions, DataFrame functionality, and various numerical algorithms	
• Archived on Zenodo, 10.5281/zenodo.10815823 software paper published in JOSS, 10.21105/joss.10366	

Puruspe	Feb. 2020 – Present
- Pure Rust library for special functions with no external dependencies - Implements gamma, beta, and error functions, including their regularized and inverse versions - Lightweight and efficient implementation ideal for mathematical and scientific computing	☆ : 29 ↓ : 1,317,072
PyTorch Template	Aug. 2024 – Present
- Flexible PyTorch template for ML experiments with modular structure - Supports YAML-based configuration for easy experiment setup and reproducibility - Supports multiple random seeds, device selection, and learning rate scheduling for robust experimentation	☆ : 10
Neural Hamilton	Sep. 2024 – Present
- Official implementation of the paper Neural Hamilton [2410.20951] - Operator-learning architectures (DeepONet, TraONet, VaRONet, MambONet) for Hamilton's equations - Rust-based generator for physically plausible potentials via Gaussian random fields and cubic B-splines	☆ : 14
Quantum Algorithms	Dec. 2023 – Jun. 2024
- Implement quantum algorithms in PennyLane, RustQIP, Qiskit and Cirq - Provide jupyter notebooks for quantum algorithms with detailed descriptions and interactive visualizations	☆ : 7

Study Groups

SYTEARK Machine Learning Study Group [SYTEARK; MLStudy, ML2022, ML2021]	2019 – Present
- Founder and organizer of a machine learning study group for theoretical physics - Current participants: 3 High energy physics graduate students, 1 postdoc and 1 A.I. researcher - Weekly rotational presentations on machine learning journals, topics, and personal research	
Yonsei Cosmo-Chalk [Yonsei_Cosmo-Chalk]	2024 – Mid 2025
- Co-organizer of the weekly blackboard talks in Yonsei HEP-COSMO. - Facilitates research presentations, journal discussions, and collaborative dialogues among group members - Fosters a supportive environment for intellectual growth and peer-to-peer learning in high-energy physics and cosmology	
ML and QML Study Group	2021 – Mid 2024
- Participated in study group led by Prof. Myung Hoon Park (Seoultech.) and Prof. Kyoungchul Kong (U. Kansas) at KIAS - Study about machine learning and quantum machine learning	

Services

Web master for Yonsei HEP-COSMO	Mar. 2017 – Aug. 2025
Developed and maintained the Yonsei HEP-COSMO website	
Beta Reader for Jpub Publishing	Oct. 2019 – Mar. 2025
- Beta reader for Jpub publishing, specializing in error detection for Rust and deep learning publications. - Reviewed a total of 16 books as a beta reader.	
Judge for SPARCS Science Hackathon 2024	Oct. 2024
SPARCS Science Hackathon 2024, “SW for Science”, held by KAIST & Daejeon Metropolitan City	
Web Assistant for the College of Science, Yonsei University	Sep. 2017 – Aug. 2022
Improved and maintained the College of Science website	